

CAPE WRATH LH, United Kingdom

WMO: 030490

Lat: 58.6253N

Lon: 4.9995W

Elev: 367

StdP: 14.50

Time Zone: 0.00 (EUW)

Period: 86-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.3	32.8	21.8	16.6	36.5	23.8	18.3	36.8	66.5	41.0	58.8	43.9	17.2	140	0.925

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	66.9	58.7	63.7	57.5	61.1	56.5	60.3	64.3	58.6	62.3	57.0	60.1	8.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.5	74.1	62.2	56.9	69.9	60.3	55.5	66.5	58.2	26.7	64.7	25.6	62.5	24.6	60.2	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 53.2	47.6	43.2	28.8	72.3	2.5	4.7	27.0	75.7	25.5	78.4	24.2	81.0	22.4	84.5		
(5)			27.1	63.5	2.0	2.9	25.6	65.6	24.4	67.3	23.3	69.0	21.9	71.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	40.8	39.8	40.8	42.9	46.9	50.3	54.2	54.6	51.8	48.4	44.4	41.5	(6)
(7)		DBStd	6.54	3.91	3.71	3.96	3.51	4.94	4.13	4.15	3.91	3.31	4.03	3.70	4.09	(7)
(8)		HDD50	1772	286	286	285	215	123	41	4	2	18	78	170	264	(8)
(9)		HDD65	6789	751	706	750	663	562	441	336	323	396	515	618	728	(9)
(10)		CDD50	460	0	0	0	1	27	50	134	146	72	28	2	0	(10)
(11)		CDD65	4	0	0	0	0	1	0	1	2	0	0	0	0	(11)
(12)		CDH74	3	0	0	0	0	0	0	0	3	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	19.0	24.5	23.4	22.0	16.6	15.4	14.1	14.9	15.5	18.3	19.6	20.7	22.7	(14)
(15)	Precipitation	PrecAvg	53.9	6.1	4.9	4.8	3.5	3.2	2.8	3.1	3.9	4.5	5.4	5.8	6.0	(15)
(16)		PrecMax	69.7	11.0	8.7	10.8	6.1	6.8	4.8	5.3	7.8	10.8	10.4	8.7	10.3	(16)
(17)		PrecMin	39.6	1.2	1.6	1.1	1.3	0.2	1.1	1.4	1.8	1.2	1.9	1.3	1.1	(17)
(18)		PrecStd	7.0	3.0	2.1	2.3	0.9	1.4	1.0	1.1	1.5	2.0	2.1	2.0	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.9	50.4	52.4	55.5	68.1	67.9	70.5	73.1	63.8	62.3	54.3	52.6	(19)
(20)		0.4%	MCWB	48.0	46.9	46.7	49.3	56.4	57.1	60.7	63.4	58.0	57.0	50.4	49.4	(20)
(21)		2%	DB	49.4	47.9	49.2	52.1	62.9	62.4	66.4	66.2	60.2	58.0	52.2	50.1	(21)
(22)		2%	MCWB	46.8	45.9	45.7	48.4	54.2	55.4	59.3	59.8	55.8	53.6	49.7	48.6	(22)
(23)		5%	DB	48.1	46.4	47.4	49.5	56.8	59.0	63.4	62.4	57.9	55.4	51.0	49.0	(23)
(24)		5%	MCWB	46.3	45.0	44.7	46.5	50.6	54.6	58.2	58.0	54.5	52.1	48.9	47.6	(24)
(25)		10%	DB	46.6	45.0	45.9	47.8	52.9	55.6	60.6	59.8	56.1	53.3	49.6	47.8	(25)
(26)		10%	MCWB	45.0	43.4	43.9	45.7	48.7	52.3	56.6	56.7	53.5	50.4	47.6	46.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	48.3	48.4	51.0	57.6	58.6	62.9	64.0	59.7	57.3	52.5	50.2	(27)
(28)		0.4%	MCDWB	50.1	49.4	50.0	54.0	65.8	63.6	68.6	71.3	61.8	60.9	53.2	51.4	(28)
(29)		2%	WB	47.9	46.6	46.6	48.9	54.3	56.3	60.1	60.7	56.9	54.5	51.0	49.1	(29)
(30)		2%	MCDWB	48.7	47.3	48.1	50.9	62.0	60.9	64.4	64.7	58.8	57.3	51.5	50.0	(30)
(31)		5%	WB	46.6	45.3	45.6	47.3	51.4	54.5	58.5	58.9	55.4	52.7	49.6	47.9	(31)
(32)		5%	MCDWB	47.5	46.1	46.8	48.8	55.4	58.3	62.6	61.9	56.8	54.5	50.4	48.8	(32)
(33)		10%	WB	45.3	43.8	44.5	46.1	49.3	52.7	56.9	56.9	54.1	51.2	48.1	46.5	(33)
(34)		10%	MCDWB	46.6	44.9	45.6	47.5	51.9	55.3	60.2	59.0	55.7	52.8	49.2	47.7	(34)
(35)	Mean Daily Temperature Range		MDBR	5.0	4.8	5.3	5.2	5.4	5.0	5.0	4.7	4.9	4.7	4.8	(35)	
(36)		5% DB	MCDBR	6.5	5.9	7.1	7.6	11.7	10.7	9.8	9.8	7.3	8.3	6.4	6.0	(36)
(37)			MCWBR	6.9	6.4	6.0	5.7	6.9	5.9	6.0	6.0	6.0	6.9	7.0	6.8	(37)
(38)			5% WB	MCDWB	5.9	5.8	6.5	6.7	10.4	9.3	8.8	9.3	5.9	7.2	5.7	5.9
(39)		MCWBR		6.7	6.6	6.2	5.7	6.8	5.8	5.9	6.3	6.0	6.9	7.1	6.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.345	0.360	0.375	0.372	0.366	0.372	0.362	0.355	0.340	0.320	0.307	(40)	
(41)		taud	2.148	2.263	2.324	2.331	2.393	2.448	2.445	2.480	2.474	2.385	2.206	2.095	(41)	
(42)		Ebn at Noon	147	211	246	263	273	276	271	266	248	213	150	111	(42)	
(43)		Edh at Noon	16	23	29	34	34	33	32	29	25	20	15	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg	100	285	628	1105	1492	1476	1366	1116	737	373	140	65	(44)	
(45)		RadStd	12	36	45	75	105	128	98	64	48	33	15	7	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (23 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page